Session Objectives

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1. List and describe the opportunistic fungal pathogens.
2. Describe the risk factors that lead to opportunistic fungal infections.
3. Describe the diseases caused by opportunistic fungal pathogens
4. Outline the laboratory techniques used for the identification and diagnosis of opportunistic fungal pathogens
5. List the treatment options for the various mycoses

Introduction

- Opportunistic mycoses
- High-risk patients:
  - Immunosuppressed state (e.g. AIDS)
- Other risk factors:
  - Large burn wounds
  - Trauma
  - Diabetes
  - Renal insufficiency requiring dialysis
  - Medical devices

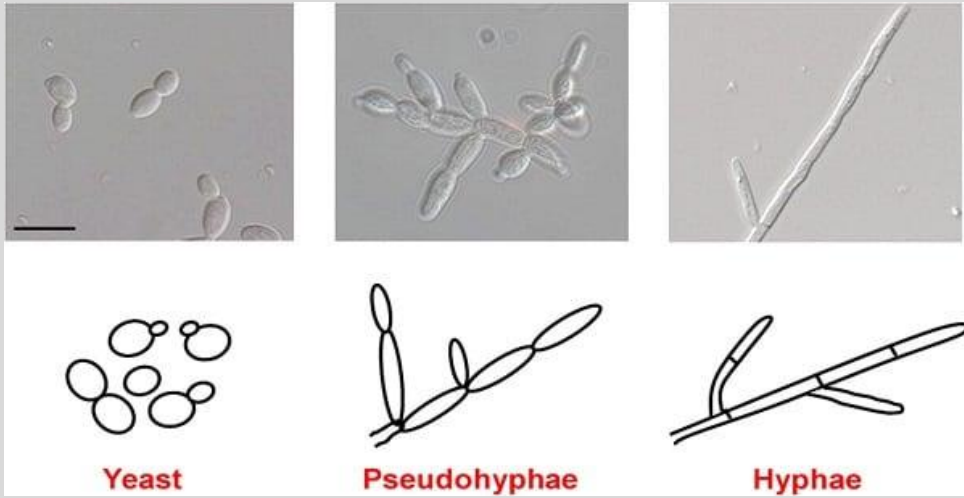
- Opportunistic fungal diseases:
  - Candidiasis
  - Aspergillosis
  - Cryptococcosis
  - Pneumocystis
  - Mucormycosis

Candidiasis

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- *Candida* spp. part of normal human flora (endogenous opportunist)
- Most prevalent systemic mycoses
- *C. albicans*
    - Others: *C. glabrata, C. auris*
- Round/oval yeasts form hyphae or pseudohyphae

- Immune defenses to prevent infection:
    - Skin & mucous membranes
    - Cell mediated immunity
    - Neutrophils



Candidiasis: Clinical Manifestations

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- **Chronic Mucocutaneous Candidiasis:**
  - Prolonged infections
  - IL-17 issues predispose

- **Disseminated candidiasis:**
  - Spreads from blood to other areas
  - Caused by:
    - Immunosuppression
    - Medical procedures and devices
    - IV drug use



Source: Usatine RP, Smith MA, Mayeaux EJ, Chumley HS: *The Color Atlas of Family Medicine, Second Edition*: www.accessmedicine.com
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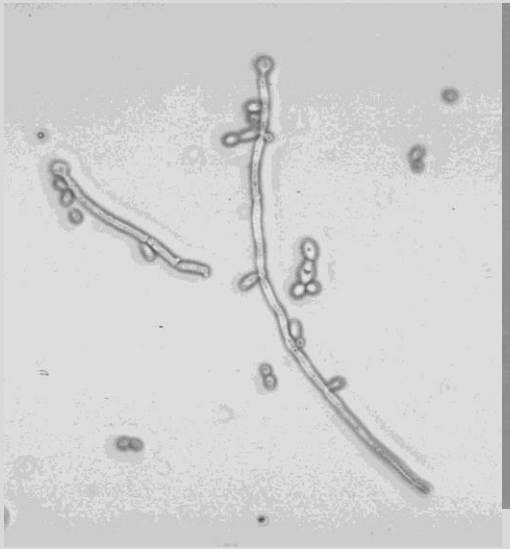
Source: D. L. Kasper, A. S. Fauci, S. L. Hauser, D. L. Longo, J. L. Jameson, J. L
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Candidiasis

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- **Diagnosis:**
  - Visualize budding yeasts, hyphae or pseudohyphae
  - *Candida albicans* forms germ tubes at 37°C
  - Sugar fermentation for non-albicans spp.
  - Culture
  - PCR

- **Treatment:**
  - Topical nystatin or azole
  - Azoles (vaginal or oral)
  - Amphotericin B with an azole or caspofungin (disseminated)

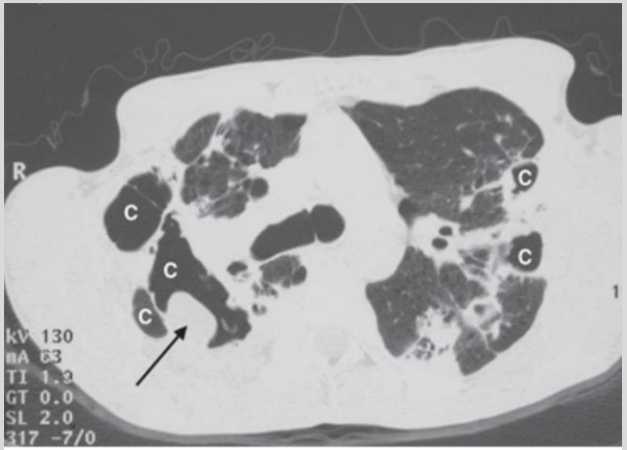




Aspergillosis: Clinical Manifestations

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- **Invasive Pulmonary Aspergillosis:**
  - Acute
    - Symptoms: fever, cough, chest discomfort, hemoptysis, shortness of breath
    - Often disseminates
  - Chronic
    - Growth in pulmonary cavities over time
    - Can display an aspergilloma (fungal ball)



Source: D. L. Kasper, A. S. Fauci, S. L. Hauser, D. L. Longo, J. L. Jameson, J.
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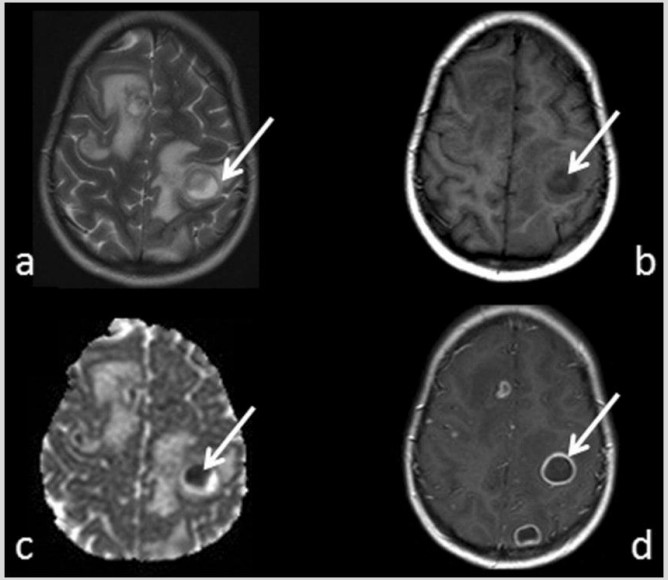
C= cavity, black arrow points to aspergilloma

Aspergillosis: Clinical Manifestations

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- **Disseminated Aspergillosis:**
  - Severely immunocompromised
  - Can spread to brain, skin, thyroid, bone, kidney, liver, GI tract, eye, & heart valves
  - Can form abscesses

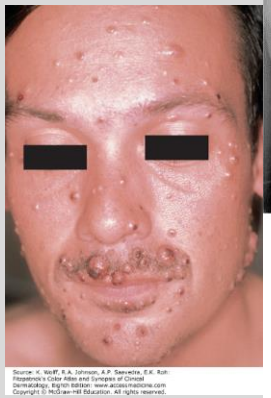
- **Allergic Bronchopulmonary Aspergillosis (ABPA):**
  - Hypersensitivity reaction to *A. fumigatus*
  - Bronchial obstruction with mucous plugs, breathlessness
  - Elevated IgE, eosinophilia

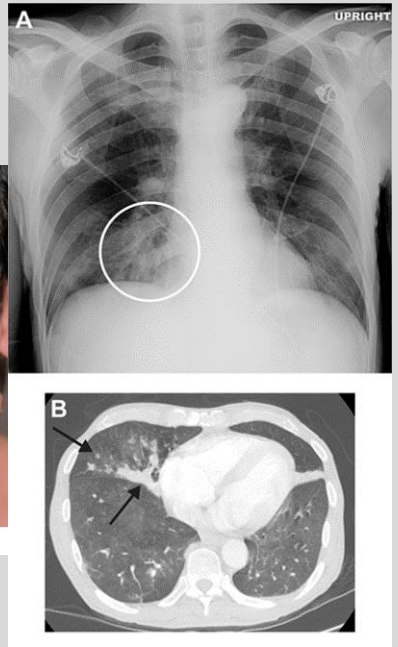


Cryptococcosis: Clinical Manifestations

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- **Pulmonary infection:**
  - Usual site of infection
  - Cough, increased sputum, chest pain, etc.
  - Most cases self resolve
- **Disseminated:**
  - Cutaneous lesions
    - Highly variable: papules, plaques, vesicles, etc.
  - Meningitis
    - Headache, fever, neurological complications
- Can form cryptococcomas (granulomatous masses)

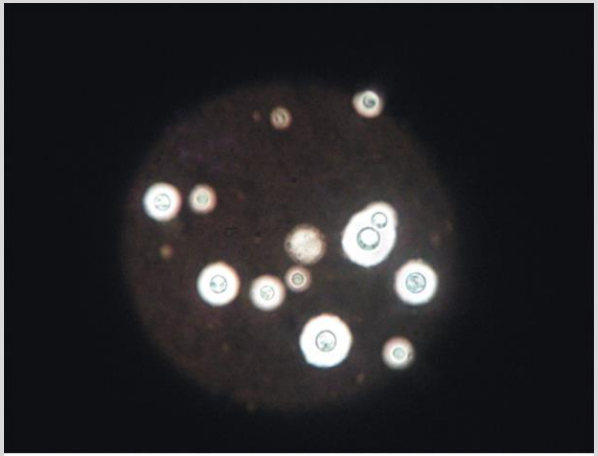




Cryptococcosis

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- **Diagnosis:**
  - Visualization in cerebrospinal fluid (CSF)
    - India ink stain reveals fungal cell with large capsule
  - CSF/blood cultures
  - Cryptococcal capsular antigen (CRAg) testing in CSF & blood -latex agglutination test
  - PCR
- **Treatment:**
  - Amphotericin B & flucytosine (disseminated/meningitis)
  - Fluconazole (long term treatment or lung infections)



Source: K.J. Knoop, L.B. Stack, A.B. Storrow, R.J. Thurman:
The Atlas of Emergency Medicine, 4th Edition,
www.accessemergencymedicine
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Mucormycosis: Clinical Manifestations

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- **Rhino-Orbital-Cerebral Disease:**
  - Non-specific sinusitis symptoms
  - Spreads to orbit, palate, facial soft tissue
  - Can form black necrotic eschar
- **Pulmonary Disease:**
  - Dyspnea, cough, hemoptysis, chest pain, fever
- **Cutaneous Disease:**
  - Invasive; necrotizing
- **Gastrointestinal Disease**
- **Disseminated Disease**





Source: Warren Levinson: Review of Medical Microbiology and Immunology, 14th Edition, www.accessmedicine.com
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Summary

Disease	Organism (most common)	Clinical Features	Diagnosis	Treatment
Candidiasis	*Candida albicans*	Mucocutaneous Can disseminate to eyes, kidney, heart, brain, liver, spleen, skin	*C. Albicans* germ tubes at 37°, Mucocutaneous=visualize yeasts, hyphae or pseudohyphae on slides, disseminated=culture	Disseminated=amphotericin B, Cutaneous=topical azoles; nystatin
Aspergillosis	*Aspergillus fumigatus*	Invasive Pulmonary Aspergillosis, Sinusitis, ABPA, disseminated	Biopsy shows septate hyphae that branch at 45°, conidia in radiating chains Culture	Voriconazole 1st line, Caspofungin, posaconazole, amphotericin B 2nd line
Cryptococcosis	*Cryptococcus neoformans*	meningitis, pulmonary disease, cutaneous disease	India ink stain reveals fungal cell with large capsule Capsular antigen testing—latex agglutination	Combined Amphotericin B and flucytosine Pulmonary disease: fluconazole
Pneumocystis	*Pneumocystis jirovecii*	Pneumocystis pneumonia, extrapulmonary symptoms rare	Diffuse interstitial pneumonia with "ground glass" infiltrations, small round cysts	trimethoprim-sulfamethoxazole (TMP-SMX) pentamidine
Mucormycosis *diabetes=unique risk factor	*Rhizopus, Mucor species*	Rhino-Orbital-Cerebral Disease, pulmonary, cutaneous, GI, disseminated	Wide, thick-walled, ribbon-like, nonseptate hyphae with branching at right angles	Amphotericin B

*immunocompromised state is a risk factor for all opportunistic mycoses

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